

DEVOPS PROJECT · EXPLAINED VISUALLY

Docker From Zero: learn it by **breaking** it

a free hands-on lab for complete beginners

DAY 1

```
$ docker --version
```

run

build

break ✨

debug 🔍

fix ✓

most tutorials stop
at "hello world"

THE END

```
$ docker compose up -d
```

THE PAIN

Why Docker beginners **get stuck**

it is not the commands. It is everything around them.



Copy, paste, hope

commands work, but
nobody explains why



Failures are hidden

the happy path only,
until your first real error



Outdated steps

PostgreSQL 18 refuses the
volume path old guides use



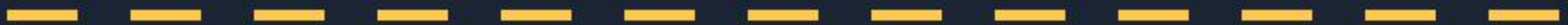
Debugging = guessing

restart it, google it,
delete everything, retry

THE IDEA

Driving school, **not a manual**

nobody learns to drive by reading. You drive, stall the car, and an instructor shows you why.



Theory

23 lessons

Instructor

guided tutorial

Practice

18 labs

Emergencies

10 breakdowns

Driving test

the capstone

stalling the car is part of the lesson

WHEN TO USE IT

Made for the **first steps**

you know a terminal. You have never run a container.



New to Docker

start at `docker --version`



Interview prep

25 questions + answers



Teaching a class

ready labs + solutions



A refresh

Docker 29 behaviour



Skip it when...

you already run containers in production every day
you want Kubernetes or CI/CD (they come next,
and they all build on what this lab teaches)

HOW YOU USE IT

Five steps, **all on your laptop**

no cloud account, nothing to pay

1

Clone the repository

only Docker + Git needed

2

Follow the tutorial

12 chapters, type every command

3

Do the lab, then break it

observe → investigate → fix → verify

4

Solve the challenges

19 tasks, solutions hidden

5

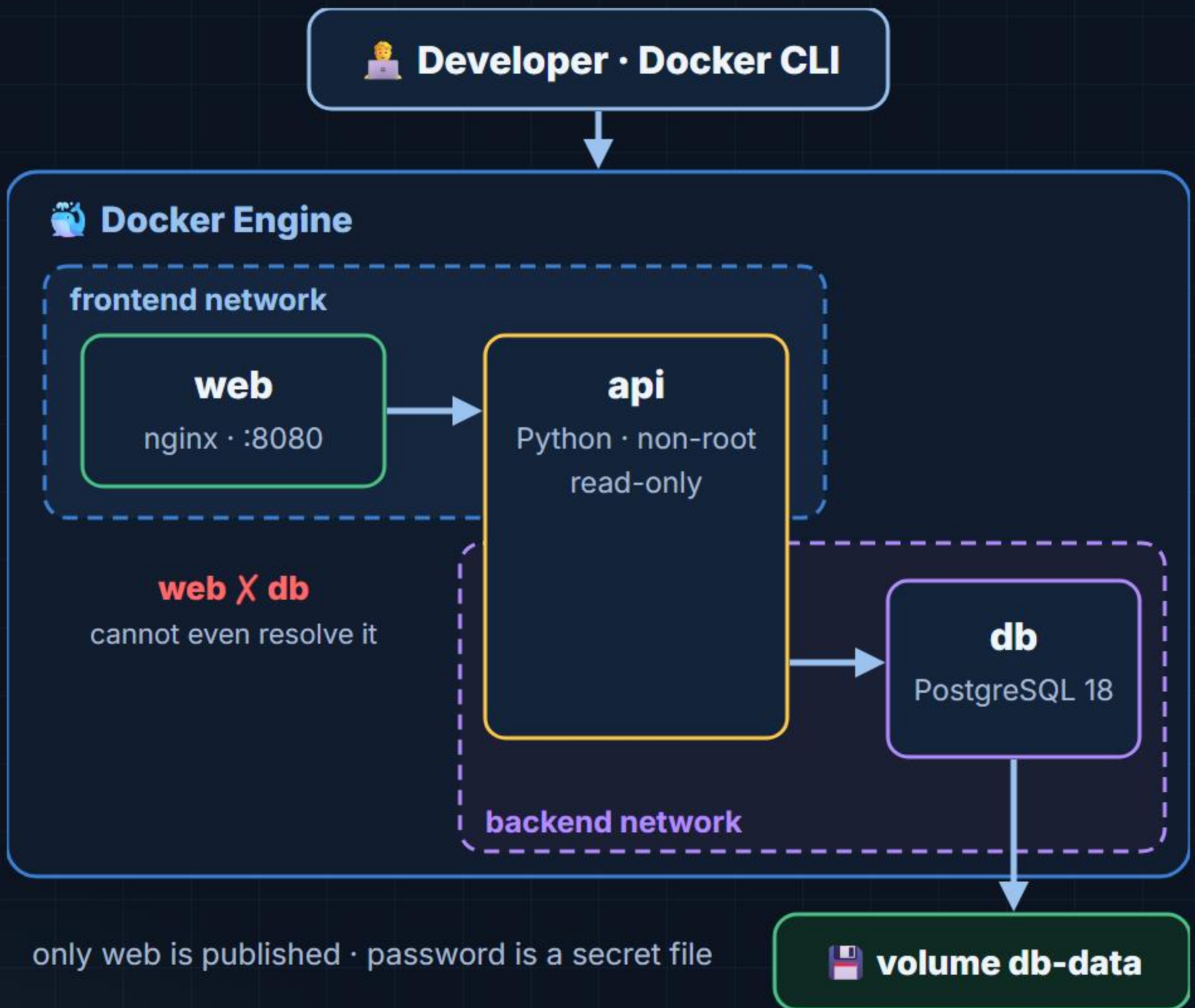
Build the capstone

then tick the knowledge checklist

THE CAPSTONE ARCHITECTURE

3 containers, 2 networks, 1 volume

built and run with one command



HOW IT WORKS

Every lesson is a loop, **every** command is tested



MEASURED RESULTS · LAB

Same app, **16x smaller** image

and real failures, reproduced on purpose

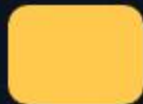
IMAGE SIZE (disk usage)

first Dockerfile



1.75 GB

slim base



212 MB

multi-stage



108 MB

FAILURES YOU WILL CAUSE (AND FIX)

Exited (2) · can't open file

port is already allocated

host not found in upstream

DB_PASSWORD missing · (3)

app bound to 127.0.0.1

OOMKilled · exit 137

capstone verify.sh: 18 checks

incl. the data surviving docker compose down + up

STUDY MATERIAL

Everything a beginner needs, in one repo

23

lessons

18

hands-on labs

12

tutorial chapters

10

failure scenarios

19

challenges

1

capstone



2-part video

recorded from real runs



PDF study guide

+ 25 interview questions

RUN IT YOURSELF

From clone to running app in 5 commands

needs only Docker and Git

```
1 $ git clone .../sufyan-devops-docker-from-zero.git
```

```
2 $ cd sufyan-devops-docker-from-zero/capstone
```

```
3 $ cp secrets/db_password.txt.example \
  secrets/db_password.txt
```

```
4 $ docker compose up -d --build
```

```
5 $ ./verify.sh
```

All checks passed.

then open <http://localhost:8080>

FREE AND OPEN SOURCE

Clone it. Break it. **Fix it.**

What was the first Docker error that confused you? 💬



The repository

[github.com/sufyanahmadkamboh/
sufyan-devops-docker-from-zero](https://github.com/sufyanahmadkamboh/sufyan-devops-docker-from-zero)



All my projects

sufyanahmadkamboh.github.io

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